



TECHNICAL NOTE — DESIGN, TESTING AND LAUNCH

Xaman Tek 2: A fully SRAD 10k ft Apogee Rocket

Solid rocket motor design for a 3km apogee rocket carrying 1U cubesat.

AUTHOR	DATE	STATUS
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This brief technical report shows the results of Xaman Tek 2, Vultur's Rocketry second rocket. Angel Vazquez, founder of the team, played the role of Lead and Propulsion Engineer during this mission.

01 Introduction

Xaman Tek - 2 (XT2) mission was to carry a 1U Cubesat payload to a 3km apogee for atmospheric sensing. This was not only the second team's rocket, but it was intended and succeeded in being the first ultra-light rocket in Mexico. With a dry mass of 13kg and a final mass of 17.3kg.

Its dual-diameter shape made it the first one in its class in Mexico and Latin America for student purposes. This geometry allowed the team to minimize aerodynamic drag and mass, while carrying a Cubesat and economizing in the manufacturing process due to the use of Cost Off The Shelf (COTS) mouldings and optimization of propellant consumption.

NOTE

Buchanan's Mission goal was to launch a 1U Cubesat in collaboration between Vultur Rocketry (VR) and Centauri Cubesat to a 3km apogee; both teams belong to Aerospace Engineering Alumni Society (IASA), founded by Angel Vazquez. The name of the mission is in honor to Walter Cross Buchanan, a pioneer mexican rocket engineer.

02 Project Management Methodology

2.1 NASA SYSTEMS ENGINEERING

2.2 SCRUM

2.3 AGILE

03 Solid Rocket Motor (SRM) Design and Testing

In addition to serving as chief engineer, Angel was directly involved in the design, manufacturing, and testing of the Solid Rocket Motor used in this mission. The VR-02 motor used a potassium nitrate / sorbitol (KNSB) propellant grain in a BATES configuration.

3.1 TSIOLKOVSKY ROCKET EQUATION AND DELTA V BUDGET

One of the particular goals of XT2 was to minimize propellant consumption while maximizing payload capability. As this was the very first VR's 3km apogee rocket, there where no previous data to start iterating over the Tsiolkovsky Rocket Equation, both of lack of dry mass and exhaust velocity values.

In that way, Angel proposed the implementation of a Neural Network (NN) for predicting the necessary propellant mass in order to reach a given apogee. This neural network might be trained with openRocket

data of the simulation of XT2 sketch and COTS SRMs.

The results of the NN must be complemented with a Delta-V budget calculator developed by Angel in Python.

$$\Delta v = v_e \ln\left(\frac{m_0}{m_f}\right)$$

(Eq. 1)

- r

propellant surface regression rate [mm/s]
- a

burn rate coefficient [mm/s · MPa⁻ⁿ]
- P_n

chamber pressure [MPa]
- n

pressure exponent [dimensionless]

2.2.1 THRUST AND SPECIFIC IMPULSE

$$F = I_{sp} \cdot \dot{m} \cdot g_0 + (P_e - P_a) \cdot A_e$$

(Eq. 2)

- F

thrust [N]
- I_{sp}

specific impulse [s]
- $\dot{m}$

mass flow rate [kg/s]
- P_e, P_a

exit / ambient pressure [Pa]
- A_e

nozzle exit area [m²]

3.1 KEY PERFORMANCE PARAMETERS

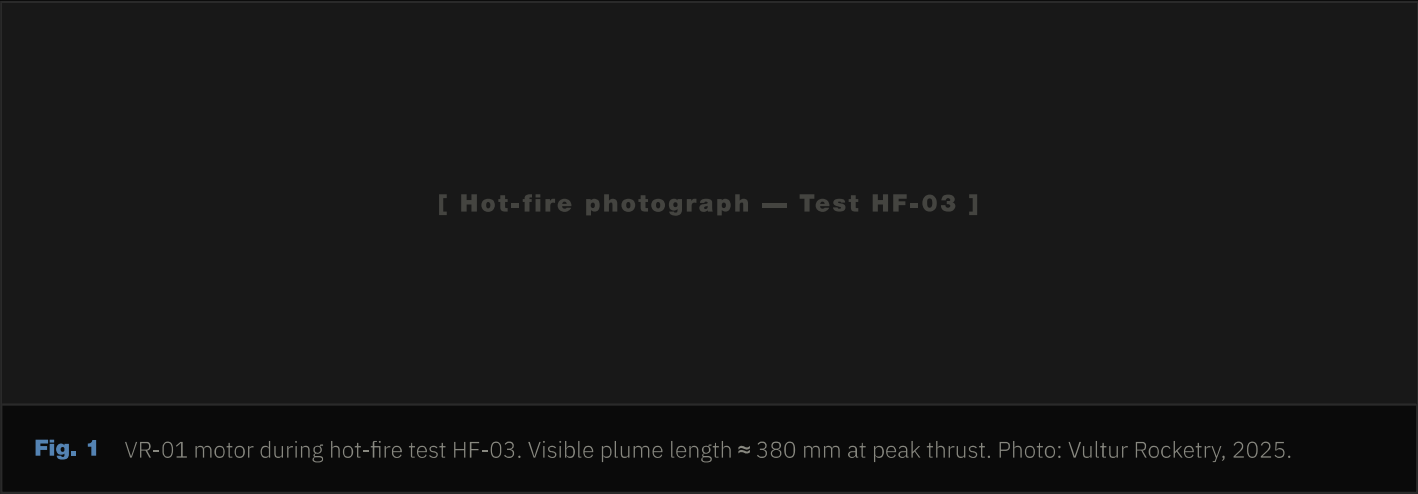
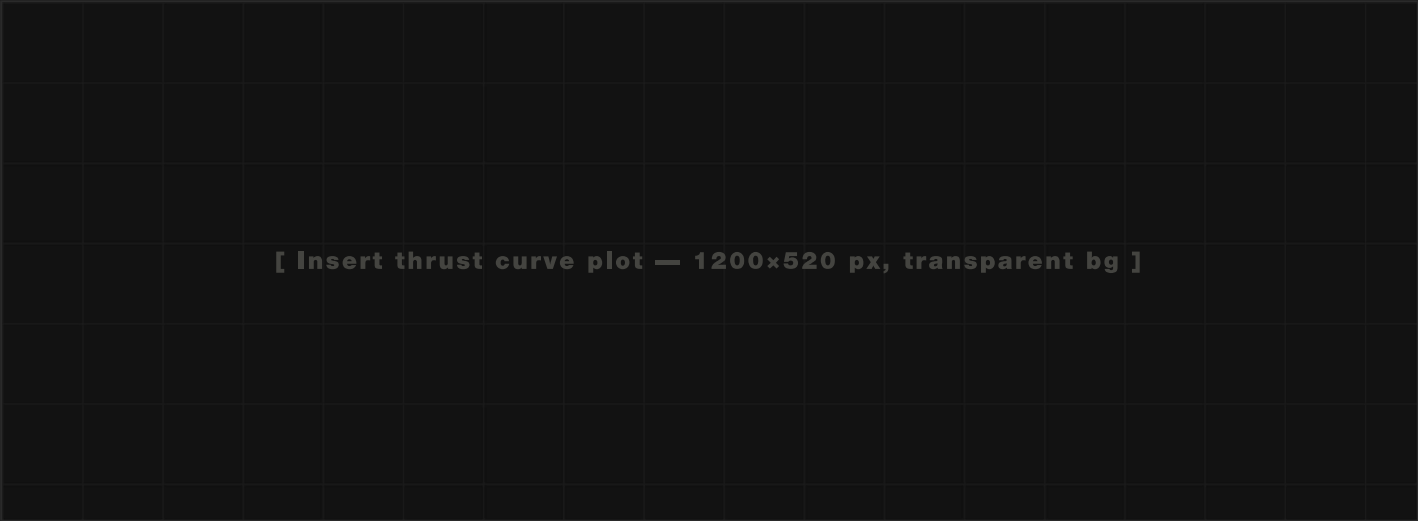
PEAK THRUST 312 N	BURN TIME 2.84 s	TOTAL IMPULSE 718 N · s
ISP (MEASURED) 163 s	AVG PC 4.2 MPa	BURN RATE (A) 8.26 mm/s · MPa⁻ⁿ

3.2 BURN RATE DATA

TEST ID	PRESSURE [MPa]	BURN RATE [MM/S]	Δ FROM MODEL [%]	STATUS
SB-01	2.10	14.3	+1.4	✓ Valid

TEST ID	PRESSURE [MPa]	BURN RATE [MM/S]	Δ FROM MODEL [%]	STATUS
SB-02	3.50	19.8	−0.8	✓ Valid
SB-03	5.00	26.1	+2.1	✓ Valid
SB-04	6.20	30.7	−3.6	⚠ Review

3.3 THRUST CURVE



04 Launch Campaign

XT2 was intended to be launched during ENMICE 2024 but it was not possible to launch during the competition as the team faced several technical issues during its development and the lack of experience in 3km rockets. Nevertheless, it was a very important experience that made the team realize they were developing a novel rocket. In that context, Vultur Rocketry kept developing XT2. During CONACES 2024, the Mexican Space Agency (AEM) asked all mexican teams to launch a 3km apogee rocket as an exhibition launch at Playa Bagdad, Matamoros. Vultur Rocketry was the only team able to launch such rocket and then, with an all-covered trip to Matamoros, XT2 was succesfully launched.

PARAMETER	SIMULATED	MEASURED	ERROR [%]
Peak thrust [N]	307	312	1.6
Burn time [s]	2.90	2.84	2.1
Total impulse [N-s]	724	718	0.8
Isp [s]	165	163	1.2

05 Conclusions

The empirical burn rate law derived from strand burner data accurately predicts VR-01 motor performance to within 2.1% across all key parameters. The pressure exponent $n = 0.32$ confirms stable burning behavior within the operational pressure range (2.0–6.5 MPa).

Recommended next steps include extending the characterization to elevated temperatures ($\pm 40\text{ }^{\circ}\text{C}$) and initiating nozzle erosion analysis for burn times exceeding 4 s.

06 References

[1]

Sutton, G.P., Biblarz, O. *Rocket Propulsion Elements* , 9th ed. Wiley, 2017.

[2]

Nakka, R. *Solid Propellant Burn Rate Determination* . Richard Nakka's Experimental Rocketry, 2019.

[3]

OpenMotor v0.5.1 — Open-source solid motor design software. github.com/reilleya/openMotor

[4]

Vultur Rocketry. *Hot-Fire Campaign HF-03 Data Package* . Internal report VR-IR-004, 2025.